

On the use of reliability test systems: A literature survey

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Abstract— Over the years the reliability test systems such as the IEEE Reliability Test System (RTS) and Roy Billinton Test System (RBTS) have been used extensively by researchers, as a bench mark system, for reliability assessment and other developments in the field of probabilistic applications in power systems. This paper presents an extensive literature survey of previous publications in which the RTS and RBTS or other test systems have been used. From the survey, several observations can be made, such as on the purpose of the use of the test systems, or where the studies were made. With the development of the electric power system, both in size and technology, this survey serves as a basis for assessing the appropriateness of the existing IEEE RTS system to address these developments. Such developments would mainly include wind energy, increased use of HVDC transmission and the state-of-the-art communication systems applied to power systems. The survey shows that it is necessary to extend the existing test systems to better address the demands of future electric power system.

Index Terms—Electric Power Systems, IEEE RTS, Probabilistic methods, RBTS, Test Systems.

I. INTRODUCTION

Reliability test systems have been used extensively in the past as a bench mark to develop and compare probabilistic reliability evaluation methods. One of the most widely used reliability test system is the IEEE RTS. The first system was published in 1979 [1]. A revised edition of the test system was published in 1986 [242] and 1996 [243]. Another widely used test system is the RBTS (Roy Billinton Test System), which was first published in 1989 [245]. The developments of these test systems were both related to the activities and persons involved in the IEEE Subcommittee of Reliability, Risk and Probabilistic Applications (RRPA). This paper is intended as the first study of an IEEE Task Force on Test Systems, led by the main author, initiated within the RRPA subcommittee with the aim to make a review of the current test systems and propose possible new developments.

There has been a large technology development in the electric power system which could be related to the traditional areas of electric power generation, transmission and consumption.

In the generation sector, the most significant changes have

been the introduction of distributed generation and more specifically, the introduction of intermittent generation from renewable energy sources.

Lately, the users of electricity become more active, which can act as both the consumers as well as the producers of electricity. This is often expressed as the new concept of *prosumer*. This could, e.g. involve the connection of electrical vehicles to the grid with the future vision that the electrical vehicles can both draw and inject (support) power from the grid. At the customer side, smart meters give the customers new possibilities to make decisions on their use of electricity. The introduction of the prosumers also result in that the distribution system needs to handle bi-directional power flow and the distribution system needs to be made more advanced, and rather similar to the transmission systems.

In the transmission area, the technology development has led to the new features with increasing use of HVDC transmission. HVDC transmissions can be used as a DC grid (e.g., for large off-shore wind farm applications) or embedded in the existing AC interconnected systems (e.g., Sweden will introduce multi-terminals voltage source converters VSC-HVDC in its high voltage transmission system to increase transfer capability and controllability of the system).

These different developments and changes in the electric power system, together with the increased use of information and communication technology (ICT), automatic control methods, etc., are reflected in the concept of Smart Grid [246].

This paper presents the first result within IEEE Task Force on Test Systems. The study gives a comprehensive review on what have been studied with the use of various test systems. The aim is to build a ground for the next step to make developments of the test systems for the use of reliability analysis for the future electric power systems – the Smart Grid.

The paper has been organized in the following way: Section II briefly describes the two popular reliability test systems. Section III gives an overview of bibliography study, Section IV provides the results of the study with respect to various aspects, and Section V concludes the paper with a few topics which can be considered as future work in current context.

II. RBTS AND RTS TEST SYSTEMS

In this Section, the two most commonly used test systems (i.e. the IEEE RTS and the RBTS) are briefly described. They have been mainly used for reliability analysis and probabilistic applications to electric power systems.

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A. IEEE RTS

The first IEEE-RTS introduced in 1979 consists of 24 buses and 32 generator units. The system is divided into two parts. The northern part has a voltage level of 230 kV and the southern part has the voltage level of 138 kV. The total installed capacity of generation units is 3405 MW and the total peak load of 2850 MW. The southern part is a power deficit area while the northern part is a power surplus area. Fig. 1 shows the original single line diagram of the IEEE-RTS (1979).

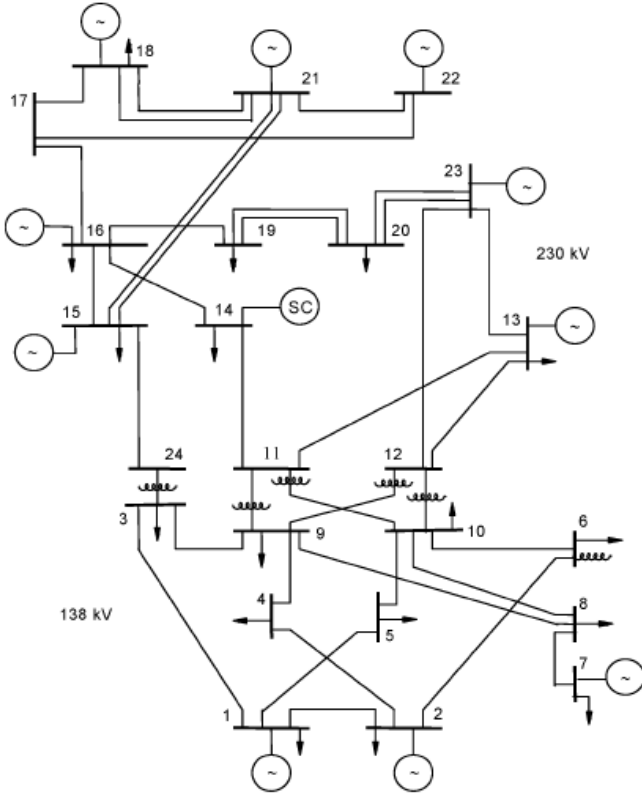


Fig. 1. IEEE-RTS 24-bus system, adopted from [162]

The RTS has been used for various kinds of studies. In recent times integration of wind power to the system has been the most studied topic in the generation sector. The RTS has been used in many cases for this kind of study. One limitation of the RTS is that no distribution system has been defined. Hence, many studies on distribution system reliability analysis have been done using self developed test system or actual distribution system connected to the RTS.

B. RBTS

The RBTS was developed at University of Saskatchewan for educational purposes and research [245]. RBTS includes 6 buses and 11 generator units. The transmission voltage is 230 kV and the total installed capacity is 240 MW with a system peak load of 185 MW. Fig. 2 shows the single line diagram of RBTS. The RBTS is a smaller test system as compared to the RTS. However, there is a distribution system defined within the RBTS. This makes it very convenient for applications on distribution system reliability analysis and similar studies. The

RBTS has also been used for a variety of similar studies as with the RTS. The area of research where it has been used the most is introducing and testing new techniques for probabilistic applications. Another advantage of the RBTS is that it is very easy to modify the test system due to its size.

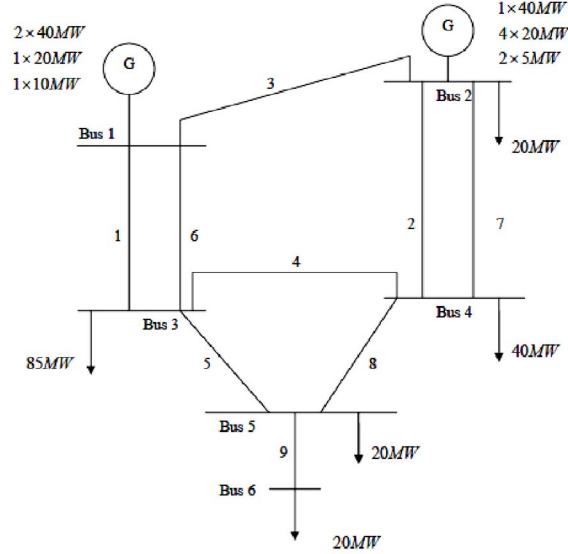


Fig. 2. RBTS 6 bus test system, adopted from [245]

III. APPROACH FOR LITERATURE SURVEY

The approach for the performed literature study can be summarized as follows:

1. Selecting journals
2. Identifying publications using test systems:
 - a) Publications using IEEE RTS or RBTS
 - b) Publications using other test systems for reliability analysis of electric power systems.
3. Various questions to be investigated:
 - a) Which test system has been used?
 - b) Who has used these test systems?
 - c) Where have the test systems have been used?
 - d) What is the purpose of using these test systems?

IV. LITERATURE SURVEY

A. Selecting Journals

In the first step, the following journals were selected for the study:

- IEEE Transactions on Energy Conversion (2010-1986)
- IEEE Transactions on Power Delivery (2010-1986)
- IEEE Transactions on Power Systems (2010-1986)
- IEEE Transactions on Smart Grid (new transaction, introduced in 2010)
- IEEE Transactions on Sustainable Energy (new transaction, introduced in 2010)

B. Selecting Publications

A total of 240 publications, as listed in Appendix 1, were selected from the journal papers listed above. The publications selected used the RTS, the RBTS or other test systems. Under this Appendix, the publications can be found in different

categories according to aspects of reliability analyses.

C. Investigating Questions

Fig. 4 shows ‘Which’ test systems have been used. It can be seen from the figure that more than half of the selected journal papers used the RTS or the RBTS for analysis. Moreover, the RTS was used with 36% of the cases. In many of these publications, modifications have been made to the RTS to fit the purpose of the study.

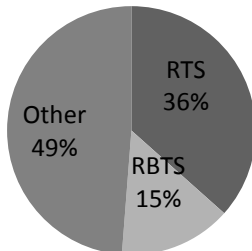


Fig. 4. General trend of usage of test systems

Fig. 5 presents results grouping the journal papers according to the affiliation of the authors. As we can see around 80% of the studies were published by universities. However the industry uses the RTS by around 20%.

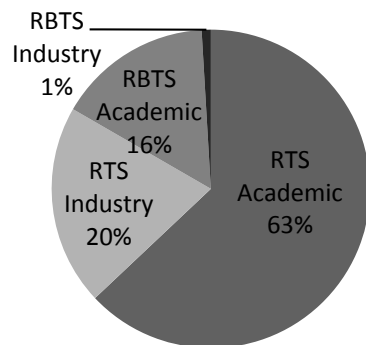


Fig. 5. The use of the RTS and the RBTS by Academics and Industry

Fig. 6 shows results from analyzing where the different studies using the test systems have been made. It can be seen that Canada and USA are the places where the RTS and the RBTS have been used the most. Due to the origin of the RBTS from Canada, this is quite expected. The main result is however that the test systems have been widely used worldwide.

The purpose behind the use of different test systems was investigated. Figure 7 and Figure 8 show the result which investigated the purpose of the studies made using the RTS and the RBTS respectively. The figures have been divided into five sections depending of different aspects of reliability analyses as follows:

- A. Reliability analysis and Planning, Generation
- B. Reliability analysis and Planning, Transmission
- C. Power System Economics and Optimization
- D. Reliability analysis and Planning, Distribution
- E. New techniques for reliability analysis

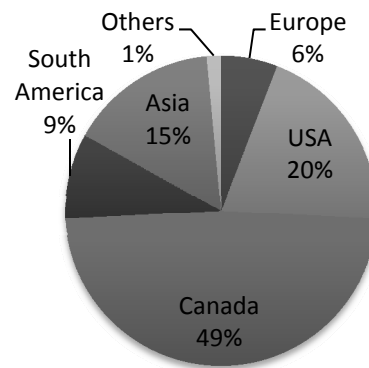


Fig. 6. Usage of RTS & RBTS worldwide

It can be seen from the Fig. 7 and Fig. 8 that the test systems have been used for a variety of studies. Fig. 7 shows that in just about half of the studies the RTS was used for analyzing the introduction of new techniques for reliability analysis. Fig. 8 shows that almost 80% of the studies have focused on reliability analysis with focus on distribution systems or introduction of new techniques. Another conclusion is that economic studies have been made more using the RTS than the RBTS system. In summary both the RTS and the RBTS have been used to analysis a wide range of questions related to reliability analysis.

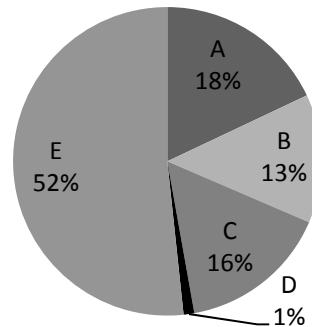


Fig. 7. The RTS used for different type of reliability studies

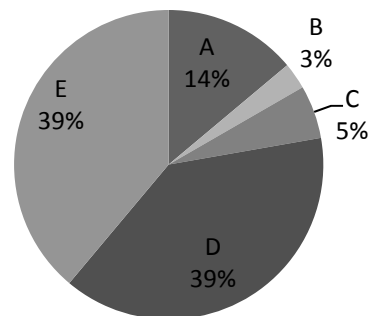


Fig. 8. The RBTS used for different type of reliability studies

V. CONCLUSIONS

This paper has presented first results from an IEEE TF on test systems, investigating further developments for reliability analysis of the future *smart* electric power systems. A bibliography study has been made identifying 240 journal

papers using test systems for reliability analysis which have been further analyzed. Of these, around 50% of the studies were made using the test systems RTS or RBTS. The study shows that these two test systems have been used worldwide and by both the industry and the academia, although almost 80% of the studies presented came from the academia. The RBTS system is more adopted for distribution system studies. Both of the systems have been used of around 50% to investigate reliability questions for distribution systems or introduction of new techniques. For analysis of economic questions, the RTS system has been more widely used than the RTS. In summary, this survey indicates the success of using the RTS and the RBTS for reliability analysis. For further work, it will be investigated how these test systems could be adopted to investigate the features for the new developments of the electric power systems, e.g. connection of large scale wind power or plug in electric cars to the grid. This work will be continued within the IEEE TF on Test System.

VI. APPENDIX: BIBLIOGRAPHY

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